LIPIDS:-

- Since we all forced to believe that eating fat will do the bad effect in our body and any direct fat will stored as fat we tend to think gat is not essential for our body.
  But that is not true anymore. Fat is the most essential nutrient for a human body.
  Fats are storage from concentrated fuel for human energy system. They are reserve store of energy too.
- Fats have poor oxygen levels and they so they are insoluble in water.
- Properties of fats are like:
- Insoluble in water
- Soluble in organic solvent such as chloroform.







CLASSIFICATION OF FATS:-

1) Simple lipids:
2) Compound lipids:





1) Simple lipids:

Simple lipids are esters of fatty acids with various glycerol and they are called triglycerides.

2) Compound lipids:

Compound lipids are simple lipids which are combined with proteins (lipoprotein) carbohydrate (glycolipids), Phosphates (phospholipids) etc..

BROADER CLASSIFICATIONS: -

1. **Saturated & Unsaturated :**
2. **Essential Fatty acids :**
3. **Trans Fatty acids :**

1) Saturated & Unsaturated :

- a saturated fat is type of fat in which the fatty acid chains have all single bonds. A fat is made two kinds of smaller molecules: glycerol and fatty acids. fats are made of long chains of carbon atoms.
- Unsaturated Fat:- an unsaturated fat is a fat or fatty acid in which there is at least one double bond within the fatty acid chain. A fatty acid chain is mono un saturated fatty acids(MUFA) contain one double bond and poly unsaturated fatty acids(PUFA) contains more than one double bond.

2) Essential Fatty acids :

- there are essential because your body cannot produce them on its own so they must come from your diet. The two primary EFAs are known as linoleic acid (Omega-6) and alpha linolenic (Omega-3). They are necessary for the following processes.
- Formation of health cell membranes.
- Proper development of functioning of the brain and nervous system.
- Proper thyroid and adrenal activity.
- Hormone production.
- Regulation of blood clotting, omega-6, encourage blood clot formation, where as, omega-3 oil reduce clotting. The ideal is to archive a balance between omega-6 and omega-3.
- Support healthy skin and hair.
- Regulation of blood pressure, liver function, immune and inflammatory responses.

3) Trans Fatty acids :

Trans fatty acids are produced when vegetable oils are hydrogenated to make margarines, vanaspati etc. Hence, major sources of trans fatty acids to human diets are commercially baked products, deep fried snacks in vanaspati and sweets. Metabolically, trans fatty acids and saturated fatty acids raise blood cholesterol levels.

SOURCES OF FATS:

Fats and Oils and their Fatty Acids

Saturated Fatty Acids	Monounsaturated Fatty Acids	Polyunsaturated Fatty Acids	
		Linoleic Acid Omega 3	Linoleic Acid Omega 6
Ghee / Butter	Olive	Low	Mustard
Coconut	Canola	Red Palm Oil	Canola
Palm Kernel	Groundnut	Palmolein	Soybean
Palm Oil	Rice Bran	Olive	Rice Bran
Vanaspati	Sesame	Medium	
	Palmolein	Groundnut	
	Red Palm Oil	Rice Bran	

FUNCTIONS OF FATS AND OILS: -

- Fats contribute to texture, flavour, taste and increases palatability of the diet. They provide and effective medium of heat transfer in deep frying and transfer of flavours from Indian spices.
- Fats have highest heat energy density of 9 kcal/g. It is the major storage form of energy in body requiring least space and minimum water for hydration. Adipose cells are 80% lipid and only 20% water and protein.
- Fats are essential for meeting nutritional needs of essential fatty acids. SFA, MUFA and cholesterol can be synthesized in the body. Excessive intake of SFA and cholesterol in diets can therefore be harmful.
- Fats promote absorption of fat soluble vitamins like vitamin A, D, E and K
- Fat intake ensures satiety. It imparts feeling of fullness and satisfaction and thus delays onset of hunger.
- Fat serve as thermal insulator in the subcutaneous tissues and certain organs. In anorexia nervosa the body hair strands erect to trap air and stimulation.
- Some dietary fats contains antioxidants which provide stability to the oil and prevent rancidity.

Excessive Fat intake: - Contemporary Indian society shows nutrition transition and access to fast food and fried snack while eating out has become a fashion. This leads to an increased intake of saturated fatty acid, milk and milk products, flesh foods and sugar at the cost of whole cereals, millets and pulses. The total calories are also high, especially if alcohol consumption is also on a rise. Hence it is important to draw the upper-limit of fat intake, as excess intake is closely linked with increased risk of developing obesity, heart diseases and cancer.

CLINICAL UNDERSTANDING OF LIPIDS:-

- Blood Lipids: the level of cholesterol and triglycerides in blood serum are frequently determined in the clinical laboratories and provide clues to the presence or absence of hyperlipidaemia.
- Triglyceride levels vary widely during the day. They are increased when chylomicrons and very low destiny lipoproteins being transported following a meal. The level increased with Lipolysis by adipose tissue as during weight loss. They decrease when fat is being synthesized by the adipose tissue or liver. The triglyceride levels increase gradually with age, but such an increase in not necessarily desirable.
- However, cholesterol and triglyceride do not exist the Free State in the circulation. Since fats are insoluble in water, protein provides the mechanism for transport in the aqueous medium of the blood. These protein- lipid complexes are known as Lipoproteins.

1) **Very low density Lipoproteins (VLDL):**
2) **Low density Lipoproteins (LDL)**:
3) **High density Lipoproteins:**
4) **Free Fatty acids (FFA):**

Very low density Lipoproteins (VLDL):

Lipoproteins are made up to cholesterol, triglycerides and proteins. They move cholesterol, triglycerides and other lipids to around the body. VLDL is one of the three main types of lipoproteins. VLDL contains the highest amount of triglycerides and a small amount of proteins.

Low density Lipoproteins (LDL):

A molecule that is combination of lipids and proteins. LDL are the chief carriers of cholesterol and are relatively low in triglycerides. When diets are rich in saturated fatty acids lipids are tranpoted in the blood. LDL transports cholesterol from the liver to the tissues of the body. LDL cholesterol is therefore considered the “bad” cholesterol.

High density Lipoproteins(HDL):

Have a protective effect ie it reduces the risk of coronary heart disease. Persons who exercise regularly, who do not smoke and are of normal weight have higher levels than those who are sedentary and obese.

Free Fatty acids (FFA):

also designated as non esterified fatty acids (NEFA) are the principal source of fatty acids that are made available to the cells for energy. The concentration of FFA in the blood at the given time is quite low but the rate of turnover is so rapid that several thousand calories are transported daily in the circulation in this way. The concentration of free fatty acids is somewhat higher in the circulation during fasting, thus indicating more rapid release from adipose tissues. It is somewhat lower when carbohydrate is being absorbed, which indicates that carbohydrate is being used for energy as well as synthesized to fat.

Liver and fat metabolism

In fat metabolism the liver cells break down triglycerides and produce energy. If someone blood sugar levels are too low, the livers break down glycogen and releases sugar into the blood. As well as sugar the liver also store vitamins and minerals(Iron and Copper) and release them into the blood when needed.

Synthesis of fat: -

Triglycerides are synthesized by the epithelial cells of the intestine mucosa, by the adipose tissue and by the liver.

Oxidation of fatty acids: --

all cells of the body except those of the central nervous system and red blood cells can oxidize fatty acids to yield energy. Although glucose is normally the only source of energy for the central nervous system, the brain cells after a period of total starvation can adapt to the utilization of ketone bodies derived from fat and amino acids.

Ketogensis: -

Ketogenesis is the biochemical process through which organisms produce. Ketone bodies through breakdown of fatty acids and ketogenic amino acids. ketosis is a metabolic state in which fat provide most of the fuel for the body. It accurse when there is limited access to glucose(Blood sugar). Which is the preferred fuel source for many cells in the body. Ketosis is most often associated with ketogenic and very low carb diets.

Cholesterol metabolism: - The liver and the intestine are the chief sites of cholesterol synthesis but all cells are able to produce some cholesterol. The endogenous production of cholesterol has been variously estimated at 800 to 1500 mg daily and is apparently independent of the dietary supply. Acetyl coenzyme A is the direct precursor of cholesterol and thus any donor of acetyl coenzyme A – fatty acids, glucose and some amino acids – is a potential source of cholesterol.

Cholesterol is transported in the blood in the various classes of lipoproteins. The body is unable to break down the cholesterol nucleus but the liver converts it by enzyme action to bile acids. This is apparently rate limited, and therefore any excess supply poses problems of disposal. Cholesterol as such and bile acids are constituents of bile and excretion occurs from the intestine.

MORDEN MYTH BUSTER:

1. Eating fat will make you fat.
2. Moderate to high cholesterol levels will hurt your heart conditions.
3. One should follow low fat diet for long time.
4. One should follow keto diet as lifestyle.
5. Body fat percentage in digital analysis machine will show perfect amount of fat.